La Scuola d'Italia / Dr. Huson / IB Math 11
27 October 2026

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2.5 Pre-Quiz: Sequences, Lines, Exponents and Radicals

A calculator is allowed. When a question asks for an exact answer or a particular form, show the algebra that leads to it.

1. In an arithmetic sequence, $u_4 = 23$ and $u_{11} = 51$.
   (a) Find the common difference d and the first term u_1.
   (b) Find an expression for u_n in terms of n.
   (c) Find S_{20}, the sum of the first 20 terms.
   (d) Show that 199 is a term of the sequence, and find which term it is.

2. In a geometric sequence, $u_2 = 12$ and $u_5 = 324$.

   (a) Find the common ratio r and the first term u_1.

   (b) Find S_8, the sum of the first 8 terms.

   (c) Find the least value of n for which $S_n > 100\,000$.

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3. The points $P(-4,\ 8)$, $Q(6,\ 3)$ and $R(4,\ 9)$ are shown on a coordinate plane. The line L passes through R and is perpendicular to the line PQ. The line L meets the line PQ at the point F.

   (a) Find the gradient of the line PQ.

   (b) Find the equation of L. Give your answer in the form $y = mx + c$.

   (c) Find the coordinates of F.

   (d) Find the distance RF. Give your answer in the form $a\sqrt{5}$, where $a \in \mathbb{Z}$.

4. At a museum, an adult ticket costs x euros and a child ticket costs y euros. A group of 3 adults and 2 children pays 61 euros. A group of 2 adults and 5 children pays 70 euros.

   (a) Write down two equations in x and y.

   (b) Find the price of an adult ticket and the price of a child ticket.

   (c) A school group of 4 adults and some children pays 156 euros. Find the number of children in the group.

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5. (a) Simplify $(3a^2b^{-1})^2 \times 2a^{-3}b^4$.

   (b) Find the exact value of $\left(\frac{16}{81}\right)^{-\frac{3}{4}}$.

   (c) Solve the equation $4^{2x-1} = 8^{x+2}$.

   (d) By letting $u = 3^x$, solve the equation $3^{2x} - 10\,(3^x) + 9 = 0$.

6. In this question, give every answer in exact form.

   (a) Simplify $\sqrt{98} - \sqrt{32} + \sqrt{18}$.

   (b) Write $\dfrac{10}{\sqrt{7} - \sqrt{2}}$ in the form $a\sqrt{7} + b\sqrt{2}$, where $a, b \in \mathbb{Z}$.

   (c) Solve the equation $x\sqrt{3} = x + 2$. Give your answer in the form $a + b\sqrt{3}$, where $a, b \in \mathbb{Z}$.

   (d) A rectangle has sides of length $\left(4 + \sqrt{5}\right)$ cm and $\left(4 - \sqrt{5}\right)$ cm. Find the area of the rectangle.